Final project - Deep Learning Course - 2025-Semester B

Avital Fine (ID. 208253823), Noa Lazar (ID. 322520339)

Submitted as final project report for the DL course, RUNI, 2025

1 Introduction

In recent years, deep learning has revolutionized the field of medical imaging, especially in automated diagnosis usingchest X-rays. Pneumonia classification is a key use case due to its high clinical importance. Traditionally, Convolutional Neural Networks (CNNs) have been the dominant architecture for computer vision tasks. More recently, Vision Transformers (ViTs) [1] have gained attention as an alternative. In this project, we compare CNN and ViT architectures for binary pneumonia detection from grayscale chest X-ray images. Our goal is to analyze their relative performance, training behavior, and generalization ability, with a focus on how well these models work on a limited dataset.

2 Methodology

Our approach focused on comparing two fundamentally different architectures for the task of pneumonia detection from chest X-ray images: Convolutional Neural Networks (CNNs) and Vision Transformers (ViTs). Both models were trained on the publicly available dataset [2], which is already split into training, validation, and test sets. We ensured that the test set remained untouched until final evaluation.

2.1 Dataset

The dataset consists of grayscale chest X-ray images, divided into two categories: "Normal" and "Pneumonia". The training set contains approximately 5,200 images, the validation set about 16 images, and the test set about 600 images. We resized all images to 224×224 pixels and normalized pixel intensities to the range $[0, 1]$.

2.2 Models

CNN: We implemented a standard CNN architecture inspired by VGG-like networks, with convolutional, pooling, and fully connected layers. This model served as a strong baseline, given CNNs' success in medical imaging. It contains approximately 404k trainable parameters. **ViT:** Our Vision Transformer implementation followed the core idea of Dosovitskiy et al. [1]. We experimented with a patch size of 16×16, embedding dimension of 256, and 6 transformer encoder layers with 8 attention heads each. This model is much larger, containing roughly 14.4M trainable parameters.

2.3 Training

Both models were trained using the Adam optimizer with a learning rate of $1e^{-4}$ and batch size of 32. Early stopping based on validation loss was employed to prevent overfitting. CNN converged within $\sim$10 epochs, while the ViT required around $\sim$15 epochs. All experiments were run locally on an Apple M4-based macOS system, with training times approximately 7 minutes for CNN and 60 minutes for ViT.

2.4 Challenges

Key challenges included: (1) preventing overfitting given the relatively small dataset, (2) ensuring fair comparison by keeping training procedures consistent across models, (3) dealing with computational efficiency and resource constraints, since the ViT required significantly more time and memory to train compared to the CNN, and (4) working with a very small validation set (16 images) compared to the test set (600 images), which made early stopping and hyperparameter tuning more difficult.

3 Experimental Results

3.1 Evaluation Metrics

We used Accuracy, Precision, Recall, and F1-Score to evaluate model performance, given the medical nature of the problem where false negatives (missed pneumonia cases) are critical.

3.2 Results

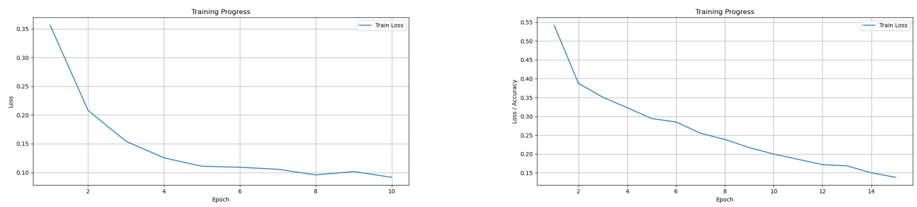
Table 1 summarizes the test set performance of CNN vs. ViT.

Model	Accuracy	Precision	Recall	F1-score
CNN	0.85	0.86	0.85	0.85
ViT	0.78	0.78	0.77	0.75

Table 1: Comparison of CNN and ViT performance on the test set.

3.3 Learning Curves

Figure 1a and Figure 1b show the training loss over epochs. CNN achieved faster convergence and outperformed ViT across all metrics, including recall. ViT required longer training and more resources but did not surpass CNN performance on this dataset.



(a) CNN Training Loss (b) ViT Training Loss

Figure 1: Training loss curves for CNN and ViT models.

4 Discussion

Our experiments revealed several important insights:

- **CNN efficiency and performance:** The CNN baseline trained significantly faster and achieved higher accuracy, precision, recall, and F1-score in fewer epochs. Its strong inductive bias for local features makes it particularly effective for detecting pneumonia patterns localized in lung regions.
- **ViT limitations on small datasets:** The Vision Transformer required more training time and computational resources. On this relatively small dataset, it underperformed compared to CNN, showing signs of overfitting due to its higher model complexity and data requirements.
- **Overfitting behavior:** CNN generalized more stably, whereas ViT tended to overfit, further highlighting the challenges of training large transformer models on limited data.
- **Model complexity:** CNN contains roughly 404k parameters, while ViT contains 14.4M, explaining the longer training time and larger memory requirements for ViT.
- **Interpretability:** ViT attention maps offered more interpretable visualizations, highlighting relevant lung regions, while CNN feature maps were less intuitive.

Overall, CNN remains the more effective and efficient model for pneumonia detection on this dataset. ViTs, however, may become competitive with larger datasets or pretraining and provide added interpretability through attention maps.

5 Code

The code for this project is available on GitHub: `deep-learning-final-project`

References

[1] Alexey Dosovitskiy, Lucas Beyer, Alexander Kolesnikov, Dirk Weissenborn, Xiaohua Zhai, Thomas Unterthiner, Mostafa Dehghani, Matthias Minderer, Georg Heigold, Sylvain Gelly, Jakob Uszkoreit, and Neil Houlsby. An image is worth 16x16 words: Transformers for image recognition at scale. 2020.

[2] Paul Mooney. Chest x-ray images (pneumonia). Available at: https://www.kaggle.com/datasets/paultimothymooney/chest-xray-pneumonia (Accessed: 2025-08-25), 2018.